

1. Which of the following statements is/are correct regarding integrated circuits (ICs)?

1 Point

- A. ICs typically contain discrete components that were fabricated on separate wafers.
- B. ICs typically contain multiple layers of metals.
- C. The transistors in an IC are typically made using Copper and Sulphur.
- D. Having multiple metal layers in an IC typically simplifies the task of routing.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only D
- ☐ only A, B
- ☐ only A, C
- ☐ only A, D
- ☐ only B, C
- ☒ only B, D
- ☐ only C, D
- ☐ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is only B, D

2. Which of the following statements is/are correct regarding photolithography?

1 Point

- A. Photolithography enables us to obtain a given geometric pattern on the surface of a silicon wafer.
- B. A photomask defines the geometric pattern we want to obtain on the silicon wafer.
- C. We use the same mask to print the patterns of all the layers in an IC.
- D. Photoresists are chemicals that are not sensitive to light.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only D
- ☒ only A, B
- ☐ only A, C
- ☐ only A, D
- ☐ only B, C
- ☐ only B, D
- ☐ only C, D
- ☐ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, B

3. Which of the following statements is/are correct regarding IC design and fabrication?

1 Point

- A. The thin slices of silicon on which an IC is fabricated are called silicon ingots.
- B. We obtain a silicon wafer after packaging a well-tested die.
- C. The primary responsibility of a fabless design company is to fabricate an IC using photolithography and other processes.
- D. ICs are typically fabricated in manufacturing plants called foundries.

- ☐ only A
- ☐ only B
- ☐ only C
- ☒ only D
- ☐ only A, B
- ☐ only A, C
- ☐ only A, D
- ☐ only B, C
- ☐ only B, D
- ☐ only C, D
- ☐ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is only D

4. Which of the following statements is/are correct regarding design styles?

1 Point

- A. We can use exactly the same VLSI design flow for full-custom, standard-cell-based, and FPGA designs.
- B. FPGAs are typically used to implement analog circuits such as PLLs, amplifiers, and ADCs.
- C. Standard-cell-based designs have lower fixed costs than FPGA designs.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☒ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is No statement is correct

5. The cost of producing N chips using standard-cell-based technology is given as:

1 Point

Cost (in Rupees)= $25,00,000+(2 \times N)$

The cost of producing N chips using FPGA technology is given as:

Cost (in Rupees)= $1,00,000+26 \times N$

The difference in the fixed cost in the two technologies and the break-even points, respectively, is:

- ☐ 1 Lakh Rupees and 50,000 chips
- ☐ 1 Lakh Rupees and 1,00,000 chips
- ☐ 10 Lakh Rupees and 50,000 chips
- ☐ 10 Lakh Rupees and 1,00,000 chips
- ☐ 12 Lakh Rupees and 50,000 chips
- ☐ 12 Lakh Rupees and 1,00,000 chips
- ☐ 24 Lakh Rupees and 50,000 chips
- ☒ 24 Lakh Rupees and 1,00,000 chips
- ☐ 25 Lakh Rupees and 50,000 chips
- ☐ 25 Lakh Rupees and 1,00,000 chips

Yes, the answer is correct. Score: 1

Accepted answer is 24 Lakh Rupees and 1,00,000 chips

6. Which of the following statements is/are correct regarding a typical VLSI design flow?

1 Point

- A. Abstraction decreases as we move from logical design to physical design.
- B. The details contained in a design decrease as we move from logical design to physical design.
- C. The turnaround time for trying out multiple solutions is typically shorter when the abstraction is higher.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☒ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, C

7. Which of the following statements is/are correct regarding hardware-software partitioning?

1 Point

- A. Hardware-software partitioning tries to exploit the distinct advantages of both hardware and software.
- B. The functionality that needs frequent updating are typically implemented using dedicated hardware rather than software running on a general-purpose processor
- C. The functionality that are the bottleneck for runtime are typically implemented using dedicated hardware rather than software running on a general-purpose processor.
- D. The cost of a product typically decreases as we implement more functionality in dedicated hardware.

- ☐ only A
- ☐ only B
- ☒ only C
- ☐ only D
- ☐ only A, B
- ☐ only A, C
- ☐ only A, D
- ☐ only B, C
- ☐ only B, D
- ☐ only C, D
- ☐ All are correct

No, the answer is incorrect. Score: 0

Accepted answer is only A, C

8. Which of the following statements is/are correct regarding RTL?

1 Point

- A. The algorithmic description of a design written in C is an example of an RTL model
- B. The RTL model describes how data moves from register to register at various clock cycles.
- C. RTL models are typically written in Hardware Design Languages (HDLs) such as Verilog and VHDL

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☒ only B, C
- ☐ All statements are correct
- ☐ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is only B, C

9. Which of the following statements is/are correct regarding SoC design methodology?

1 Point

- A. We write RTL for all subsystems from scratch in an SoC design methodology.
- B. Pre-designed IPs are not allowed in an SoC design methodology due to the high effort required for their integration.
- C. Once an IP is integrated into an SoC, we do not need to verify the SoC at the top level because the IPs were already verified by the IP vendor.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only A, B
- ☐ only A, C
- ☐ only B, C
- ☐ All statements are correct
- ☒ No statement is correct

Yes, the answer is correct. Score: 1

Accepted answer is No statement is correct

10. Which of the following statements is/are correct regarding behavior synthesis?

1 Point

- A. Behavior synthesis involves creating an RTL model (in a language such as Verilog) from an algorithmic description (in a language such as C).
- B. Behavior synthesis instantiates transistors in the layout and makes connections.
- C. Behavior synthesis typically ignores all constraints of resource usage, latency, throughput, area, and power.
- D. Behavior synthesis can reduce area by sharing resources and trading off latency.

- ☐ only A
- ☐ only B
- ☐ only C
- ☐ only D
- ☐ only A, B
- ☐ only A, C
- ☒ only A, D
- ☐ only B, C
- ☐ only B, D
- ☐ only C, D
- ☐ All are correct

Yes, the answer is correct. Score: 1

Accepted answer is only A, D